

A Multimodal Fake Job Posting Detection

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Abstract—Online job websites have become a focal point for fraudsters using internet scams to deceive job seekers by creating fictitious job listings that they have posted online. Most of the current methods used to detect these scams rely exclusively on textual features. These methods do not account for the visual inconsistencies that are used by fraudsters in the posting of these scams. To overcome this issue, this study proposes a multimodal framework for the detection of fake job postings that uses webpage screenshots along with the text content of the postings. We propose three methodologies for detecting fake job postings: (1) deep multimodal fusion with adaptive multi-task learning that optimizes the text-based and visual features of the postings using learnable loss weights, (2) simplified end-to-end deep learning that does not require auxiliary supervision, and (3) lightweight traditional machine learning with semantic consistency scoring from optical character recognition images. We have developed an automated pipeline based on the Selenium web browser to obtain screenshots of the webpages for the Electric Motor Services Company directory. The results of our experiments indicate that the addition of the visual features from the screenshots will lead to significant improvements in fake job posting detection. Our adaptive methodology outperformed the classical approach in terms of accuracy while the classical approach provided 10–20 times faster inference than our adaptive methodology.

Index Terms—Fake Job Posting Detection, Multimodal Learning, OCR, Semantic Consistency, BERT, Multi-Kernel ResNet, Adaptive Multi-Task Learning, Fraud Detection

I. INTRODUCTION

Online job boards have revolutionized the job market by providing previously unheard-of access to opportunities across the globe. Cybercriminals, however, now have new attack avenues thanks to this digital transformation. They take advantage of these platforms by creating phony job postings with the intention of obtaining personal information and financial gain from susceptible job seekers. According to studies, fake postings make up about 5% of online listings, and the repercussions go beyond monetary losses to include psychological distress and a decline in trust in reputable platforms.

Textual analysis of job descriptions is the sole focus of traditional detection methods. On the EMSCAD dataset, recent research by Srikanth et al. [1] used ensemble techniques (Bagging, XGBoost, AdaBoost) with advanced text featurization methods to achieve 98.85% accuracy. In order to detect fraud, these methods take advantage of linguistic patterns like ambiguous descriptions, impractical promises, and grammatical errors. However, textual analysis provides an incomplete

picture. Fraudulent postings often exhibit distinctive visual characteristics: unprofessional layouts, inconsistent branding, poor-quality graphics, and discrepancies between database content and webpage rendering. These visual cues are inaccessible to text-only models, representing a critical blind spot. Sophisticated fraudsters may craft convincing descriptions while revealing deception through careless visual presentation, or conversely, create polished pages that contradict database information through client-side manipulation.

A. Motivation and Contributions

We focus on three shortcomings in existing systems: limited deployment flexibility, mismatched text between database and webpage, and absent visual cues. Our contributions are:

- 1) Multimodal dataset with Selenium-based webpage screenshots for EMSCAD.
- 2) Adaptive multi-task model using TF-IDF, BoW, BERT, and Multi-Kernel ResNet features with learnable loss weights.
- 3) Multi-Kernel ResNet with parallel 3, 5, 7 convolutions for webpage screenshot analysis.
- 4) Semantic consistency score using Sentence-BERT on database text and OCR text.
- 5) Three deployment modes covering high-accuracy, balanced, and lightweight setups.

II. LITERATURE REVIEW

The proliferation of online recruitment platforms and the growing sophistication of scam operations have made the detection of fraudulent job postings a crucial research area. This section examines recent developments in the use of deep learning and machine learning methods to spot fraudulent job postings.

A. Deep Learning Approaches for Fraud Detection

A thorough investigation of online recruitment fraud detection using deep learning techniques is presented in recent work by [2]. In order to solve this issue, the authors discuss current scam trends and investigate various machine learning and natural language processing techniques. This work's discussion of useful implementation features, such as incorporating employer metadata and behavioral patterns into the detection pipeline, is a noteworthy strength. Their multi-model comparison offers insightful information about the top

architectures for textual fraud detection. Nevertheless, the study is restricted to text-only analysis, ignoring visual cues from images and page layouts that could improve detection precision. Furthermore, it's still unclear how generalizable it is across various recruitment platforms.

B. Feature Engineering and Classical ML Methods

In [3], the widely used Kaggle job-post dataset is used to present systematic feature engineering approaches for the application of traditional machine learning techniques to job posting classification. This work shows that competitive performance can be achieved with well-designed features without the need for intricate deep learning architectures. For comparative research, the methodical approach offers a repeatable baseline. However, the study is limited by its use of a single dataset, which might not adequately reflect how fraud strategies are changing in 2025. Its applicability to contemporary job postings, which often include images, logos, and rich formatting, is further limited by the lack of multimodal inputs.

C. Platform-Specific Detection Systems

One of the most well-known professional networking sites, LinkedIn, is the subject of a platform-specific analysis by [4]. Because it creates an end-to-end pipeline that includes data scraping, preprocessing, model training, and deployment considerations specific to LinkedIn's ecosystem, this research is especially valuable. Optimization based on LinkedIn's distinctive features and user behavior is made possible by the platform-specific focus. However, the reduced applicability to other platforms is the price of this specialization. Without utilizing more potent contextual models like transformers, the methodology mainly relies on traditional machine learning techniques. Furthermore, no visual components or profile screenshots that might offer more fraud indicators are included in the study.

D. Hybrid and Multimodal Approaches

[5] suggests a hybrid machine learning framework that combines textual content with numerical and metadata features for fake job posting detection in order to overcome some of the drawbacks of text-only approaches. In comparison to models utilizing single feature types, the authors show that this multimodal approach at the feature level achieves superior precision and recall. The system can find discrepancies that pure text analysis might overlook by combining structured data (like location, employment type, and salary ranges) with unstructured text. Despite these advances, the work remains unimodal at the content level, not incorporating webpage visual evidence or optical character recognition from images. Since fraudulent postings frequently display unique visual patterns in layout, logo usage, and design quality, the lack of visual analysis constitutes a substantial gap.

E. Research Gaps and Future Directions

The literature review identifies a number of recurring flaws in the fraud detection systems in use today. First, despite

the rich visual information found in contemporary job advertisements, the majority of approaches only consider textual content, ignoring visual and multimodal signals. Second, most research is limited to single datasets or particular platforms, leaving generalization across platforms understudied. Third, adversarial robustness—the capacity of models to resist sophisticated fraudsters who might modify their strategies to avoid detection systems—is not well discussed. Lastly, older datasets may overlook new scam trends, and the temporal component of fraud evolution is not sufficiently taken into account.

Truly multimodal architectures that combine text, images, metadata, and behavioral signals using deep learning frameworks like vision-language transformers should be the focus of future research. Cross-platform generalization studies would enhance the practical deployment of these systems in diverse recruitment ecosystems. In addition, incorporating adversarial training and continuous learning mechanisms would improve resilience against evolving fraud tactics.

III. MULTIMODAL EMSCAD DATASET

A publicly accessible dataset called the Employment Scam Aegis Dataset (EMSCAD) was created to identify phony job postings. We expanded the original EMSCAD dataset to create a multimodal version by adding visual data from job posting webpages, even though it contained textual information like job descriptions, company profiles, and employment details.

We used web scraping methods with Selenium, a browser automation framework, to build this multimodal extension. We methodically took screenshots of the matching webpage for every job posting in the dataset while maintaining the posting's visual layout, design components, and presentation style. A collection of webpage images that enhance the current textual elements were produced as a result of this process.

By combining textual and visual modalities, the resulting Multimodal EMSCAD dataset [6] allows for a more thorough examination of the authenticity of job postings. Additional indicators to differentiate genuine job postings from fraudulent ones include visual elements like page design quality, branding consistency, and professional appearance. By taking into account both the written content and the visual presentation of the advertisement, this multimodal approach mimics how human users naturally evaluate job postings.

IV. WORK DONE

We expanded the fake job posting detection framework to include multimodal data, building on the baseline work by Srikanth et al. [1], which used ensemble techniques on text-only features. In contrast to the baseline method, which only used textual features taken from job descriptions, our work adds webpage screenshot analysis as an extra modality. We created a thorough multimodal corpus by methodically capturing and archiving webpage screenshots for every job posting using an internal dataset enhancement pipeline built with Selenium WebDriver. This section describes three different

methods that gradually combine textual and visual elements using various architectural paradigms and learning techniques.

A. Dataset Preparation and Augmentation

We describe our multimodal dataset construction methodology before outlining the three approaches. We created an automated web scraping pipeline using Selenium WebDriver to obtain real webpage renderings of every job posting, starting with the publicly accessible EMSCAD dataset that contains textual job posting information. We programmatically recreated the job posting webpage for every record in the dataset and took full-page screenshots at a standard resolution of 1920 x 1080 pixels. A hierarchical directory structure was used to arrange these screenshots: `/images/{fraudulent}/{job_id}.png`, where `fraudulent` denotes the ground truth label (0 for legitimate, 1 for fraudulent). Seven important fields from each posting were concatenated for textual data preprocessing: title, location, department, company profile, description, requirements, and benefits. To ensure uniform text representation across all samples, missing values were substituted with empty strings prior to concatenation. All three of our approaches' text-based feature extractors use this unified text representation as input.

B. Approach 1: Deep Multimodal Fusion with Adaptive Multi-Task Learning

In the first method, joint representation learning is carried out across several heterogeneous feature spaces using an advanced end-to-end deep learning architecture. This method is intended to simultaneously learn from visual webpage characteristics and extract complementary information from both conventional NLP features and contemporary neural representations.

1) *Multi-Stream Feature Extraction Architecture*: The model architecture consists of four parallel feature extraction pipelines, each specialized for a different representation:

a) *Traditional Text Features*: We use two traditional NLP feature extraction techniques that work well for text classification tasks:

- **TF-IDF Vectorization**: From the concatenated text fields, we extract 1000-dimensional term frequency-inverse document frequency vectors with bigram support (`ngram_range=[1,2]`). The TF-IDF representation provides a statistical measure of term relevance by capturing the significance of words in relation to the entire corpus. The model can identify basic phrasal patterns that could be signs of fraudulent language by incorporating bigrams.
- **Features of the Bag-of-Words**: Term presence/absence patterns are captured by computing binary 500-dimensional BoW vectors. The binary BoW representation, which treats all present terms equally in contrast to TF-IDF, can be useful for identifying particular keywords or phrases that frequently occur in fraudulent postings, regardless of their frequency. This complementary depiction offers a different statistical perspective on the text.

To avoid data leakage, both conventional features are calculated on the training set and uniformly transformed on the test set. The learned vocabulary and IDF weights of the TF-IDF and BoW vectorizers are frozen during test evaluation, and they are only fitted on training data.

b) *Neural Text Embeddings*: For deep semantic understanding of job posting text, we utilize:

- **BERT Contextualized Embeddings**: We employ the pre-trained BERT-base-uncased model to generate 768-dimensional contextual embeddings. Input text is tokenized using the BERT tokenizer with a maximum sequence length of 256 tokens, applying truncation and padding as necessary. The [CLS] token's pooler output from the final BERT layer serves as the document-level representation. Critically, all BERT parameters are frozen (`requires_grad=False`) during training, treating BERT purely as a feature extractor rather than fine-tuning it. This design choice reduces computational requirements while still leveraging BERT's pre-trained linguistic knowledge. The BERT embeddings capture semantic relationships and contextual nuances that traditional bag-of-words methods cannot represent.

c) *Visual Feature Extraction*: For processing webpage screenshots, we developed a custom convolutional neural network architecture:

- **Multi-Kernel ResNet**: We designed a novel CNN architecture specifically for webpage analysis that extracts 512-dimensional image embeddings. The network begins with a standard ResNet-style stem consisting of a 7x7 convolution (stride 2), batch normalization, ReLU activation, and 3x3 max pooling (stride 2) to downsample the input from 224x224 to 56x56 feature maps. The core innovation lies in our Multi-Kernel Block design, which replaces standard ResNet bottleneck blocks. Each Multi-Kernel Block applies three parallel convolution branches with kernel sizes 3x3, 5x5, and 7x7 to the same input. This multi-scale processing allows the network to simultaneously capture fine-grained details (3x3), medium-range patterns (5x5), and broader contextual information (7x7) from the webpage screenshots. The three branches produce $c/3$, $c/3$, and $c/3$ output channels respectively (where c is the desired output channel count), which are concatenated along the channel dimension to produce c total channels. Batch normalization is applied to the concatenated features, and a residual connection is added from the input (with 1x1 convolution projection if channel dimensions differ), followed by ReLU activation. The complete architecture consists of four stages: layer 1 (64→128 channels, followed by 2x downsampling), layer 2 (128→256 channels, 2x downsampling), layer 3 (256→512 channels, 2x downsampling), and layer 4 (512→512 channels, no downsampling). Global adaptive average pooling reduces the final feature maps to 1x1 spatial dimensions, and flattening produces the 512-dimensional visual embedding. Unlike BERT, the ResNet

parameters are trainable, allowing the model to learn webpage-specific visual patterns relevant to fraud detection.

All images undergo preprocessing before entering the ResNet. During training, data augmentation is applied including random horizontal flips ($p=0.3$), random rotation (± 10 degrees), and color jittering (brightness=0.3, contrast=0.3, saturation=0.2). Both training and testing images are resized to 224×224 pixels and normalized using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]).

2) *Feature Projection and Fusion*: Each of the four feature streams passes through dedicated fully-connected projection layers that map them to intermediate dimensional spaces:

- TF-IDF features (1000D) $\rightarrow$ 256D via FC $\rightarrow$ ReLU $\rightarrow$ BatchNorm $\rightarrow$ Dropout(0.4)
- BoW features (500D) $\rightarrow$ 128D via FC $\rightarrow$ ReLU $\rightarrow$ BatchNorm $\rightarrow$ Dropout(0.4)
- BERT embeddings (768D) $\rightarrow$ 256D via FC $\rightarrow$ ReLU $\rightarrow$ BatchNorm $\rightarrow$ Dropout(0.4)
- Image features (512D) $\rightarrow$ 256D via FC $\rightarrow$ ReLU $\rightarrow$ BatchNorm $\rightarrow$ Dropout(0.4)

The projection layers serve multiple purposes: they normalize the different feature modalities to comparable scales, introduce non-linearity through ReLU activation, provide regularization through 40% dropout, and reduce dimensionality to make the subsequent fusion more tractable. Batch normalization stabilizes training by normalizing activations across mini-batches.

The four projected representations are concatenated along the feature dimension to form a 896-dimensional joint embedding vector ($256+128+256+256=896$). This concatenated representation is then processed by a three-layer fusion network:

- Layer 1: 896 $\rightarrow$ 512 dimensions (FC $\rightarrow$ ReLU $\rightarrow$ BatchNorm $\rightarrow$ Dropout(0.4))
- Layer 2: 512 $\rightarrow$ 256 dimensions (FC $\rightarrow$ ReLU $\rightarrow$ BatchNorm $\rightarrow$ Dropout(0.4))
- Layer 3: 256 $\rightarrow$ 2 dimensions (FC, no activation, logits for binary classification)

The fusion network progressively combines and refines the multimodal features, learning complex cross-modal interactions that are predictive of fraudulent behavior.

3) *Adaptive Multi-Task Learning Framework*: A key innovation in this approach is the adaptive multi-task loss mechanism. Rather than relying solely on the fusion classifier, we introduce auxiliary classification heads that operate directly on each projected feature representation:

- TF-IDF auxiliary classifier: 256D $\rightarrow$ 2 logits (single linear layer)
- BoW auxiliary classifier: 128D $\rightarrow$ 2 logits (single linear layer)
- BERT auxiliary classifier: 256D $\rightarrow$ 2 logits (single linear layer)
- Image auxiliary classifier: 256D $\rightarrow$ 2 logits (single linear layer)

Each auxiliary classifier attempts to predict the fraudulent/legitimate label using only its corresponding modality. This multi-task formulation serves two purposes: (1) it provides additional supervision signals that help train the feature projection layers more effectively, and (2) it enables the model to learn which modalities are most discriminative for this task.

The total loss function combines the primary fusion loss with the four auxiliary losses using learnable weight parameters:

$$\mathcal{L}_{total} = \mathcal{L}_{fusion} + \frac{e^\alpha}{\sum} \mathcal{L}_{tfidf} + \frac{e^\beta}{\sum} \mathcal{L}_{bow} + \frac{e^\gamma}{\sum} \mathcal{L}_{bert} + \frac{e^\delta}{\sum} \mathcal{L}_{img} \quad (1)$$

where $\sum = e^\alpha + e^\beta + e^\gamma + e^\delta$ normalizes the weights to sum to a constant, and $\{\alpha, \beta, \gamma, \delta\}$ are learnable scalar parameters (initialized to 1.0). Each $\mathcal{L}_i$ is computed as the cross-entropy loss between predicted logits and ground truth labels. The exponential transformation ensures positive weights, and the normalization prevents the auxiliary losses from dominating the total loss. These weight parameters are updated via gradient descent alongside the model parameters, allowing the network to automatically discover the relative importance of each modality during training.

4) *Training Configuration*: The model employs a sophisticated training strategy to handle the multi-component architecture:

a) *Optimization*: We use the AdamW optimizer with differentiated learning rates for different components:

- ResNet backbone: $1e-5$ (fine-tuning pretrained features slowly)
- BERT projection layer: $1e-4$ (adapting frozen BERT features)
- TF-IDF and BoW projection layers: $3e-4$ (learning from scratch)
- Image projection layer: $1e-4$ (complementing ResNet features)
- Fusion network: $2e-4$ (learning cross-modal interactions)
- Auxiliary classifiers: $2e-4$ to $3e-4$ (matching projection layers)
- Adaptive loss weights $\{\alpha, \beta, \gamma, \delta\}$: $1e-3$ (faster adaptation)

These differentiated learning rates reflect the different initialization states of model components. L2 weight decay of 0.01 is applied to all parameters for regularization.

b) *Loss Weighting and Regularization*: To address the severe class imbalance in the EMSCAD dataset (fraudulent posts comprise less than 5% of total samples), we compute class weights using scikit-learn's balanced class weight computation:

$$w_i = \frac{n_{samples}}{n_{classes} \times n_{samples,i}} \quad (2)$$

These class weights are incorporated into all cross-entropy loss calculations, penalizing misclassification of minority class samples more heavily. Additionally, gradient clipping with

maximum norm of 1.0 prevents exploding gradients during backpropagation through the deep multi-branch architecture.

c) *Learning Rate Scheduling*: We employ ReduceLROnPlateau scheduler that monitors validation loss. When validation loss fails to decrease for 3 consecutive epochs (patience=3), the learning rate for all parameter groups is multiplied by 0.5 (factor=0.5). This adaptive scheduling allows the model to escape plateaus and converge to better local minima.

d) *Early Stopping*: To prevent overfitting, we implement early stopping with patience of 7 epochs. The early stopping criterion monitors both validation loss and validation F1-score: training terminates if validation loss fails to decrease by at least 0.001 (delta=0.001) AND F1-score fails to improve for 7 consecutive epochs. The model checkpoint with the best validation F1-score is retained.

e) *Data Handling*: Due to computational constraints, we train on a random 20% subsample of the full dataset (random_state=42 for reproducibility), stratified by the fraudulent label to maintain class distribution. The data is split 80-20 into training and test sets using stratified sampling. Mini-batches of size 16 are used during training (shuffled) and evaluation (sequential). Training proceeds for a maximum of 50 epochs unless early stopping is triggered.

C. Approach 2: Simplified Deep Multimodal Learning

The second approach maintains the deep learning architecture from Approach 1 but simplifies the training procedure by removing the adaptive multi-task loss mechanism. This variant explores whether the added complexity of auxiliary classifiers and learnable loss weights provides substantial benefit, or whether a simpler training objective suffices.

1) *Architectural Consistency*: All architectural components remain identical to Approach 1:

- Four parallel feature extractors (TF-IDF, BoW, frozen BERT, trainable Multi-Kernel ResNet)
- Identical projection layers mapping to intermediate dimensions (TF-IDF→256, BoW→128, BERT→256, Image→256)
- Same 896-dimensional concatenated feature vector
- Identical three-layer fusion network (896→512→256→2)

The preprocessing pipeline, data augmentation strategy, and image transformations are also unchanged. This architectural consistency ensures that performance differences between Approach 1 and Approach 2 can be attributed solely to the training methodology rather than model capacity or feature engineering choices.

2) *Simplified Loss Function*: The key modification is the elimination of auxiliary classification heads and learnable weight parameters. Instead of the complex multi-task formulation, Approach 2 uses a single supervised objective:

$$\mathcal{L} = \text{CrossEntropyLoss}(y_{\text{fusion}}, y_{\text{true}}, \text{weight} = w_{\text{class}}) \quad (3)$$

where y_{fusion} are the 2-dimensional logits from the fusion network, y_{true} are the ground truth labels, and w_{class} are the

balanced class weights. The four feature extraction streams (TF-IDF, BoW, BERT, Image) are now trained end-to-end purely through the fusion classifier’s gradient signal, without explicit supervision at intermediate layers.

This simplification offers several potential advantages: (1) fewer trainable parameters (eliminates four auxiliary classifiers and four weight parameters), (2) simpler gradient flow during backpropagation, (3) faster training iterations, and (4) reduced risk of auxiliary tasks interfering with the primary classification objective. The trade-off is the loss of explicit modality-specific supervision that might help individual feature extractors learn more discriminative representations.

3) *Training Configuration*: The training configuration closely mirrors Approach 1 with minor adjustments:

- **Optimizer**: AdamW with identical layer-wise learning rates (ResNet: 1e-5, BERT projection: 1e-4, traditional feature projections: 3e-4, image projection: 1e-4, fusion network: 2e-4). Weight decay remains 0.01.
- **Class Balancing**: Same balanced class weights applied to the single cross-entropy loss.
- **Regularization**: Gradient clipping (max norm=1.0) and 40% dropout in projection/fusion layers.
- **Scheduling**: ReduceLROnPlateau with identical hyperparameters (mode='min', factor=0.5, patience=3).
- **Early Stopping**: Same patience=7 criterion monitoring validation loss and F1-score.
- **Data**: 20% dataset subsample, 80-20 train-test split, batch size=16, max 50 epochs.

The only parameter group removed from the optimizer is the adaptive loss weights $\{\alpha, \beta, \gamma, \delta\}$, as they no longer exist in this simplified formulation. This approach serves as an ablation study to quantify the value added by the adaptive multi-task learning framework versus a standard end-to-end multimodal fusion approach.

D. Approach 3: Lightweight Multimodal ML with Semantic Consistency Analysis

The third approach adopts a fundamentally different paradigm, replacing deep learning with classical machine learning algorithms augmented by carefully engineered multimodal features. This methodology emphasizes interpretability, computational efficiency, and rapid prototyping while still leveraging both textual and visual information. The core insight is that inconsistencies between structured database text and actual webpage rendering may serve as a powerful fraud indicator.

1) *OCR-Based Semantic Consistency Feature*: We introduce a novel semantic consistency score that quantifies the alignment between database text and visual webpage content:

a) *OCR Text Extraction*: For each job posting’s webpage screenshot, we apply Optical Character Recognition using the EasyOCR library (English language model, GPU acceleration when available). EasyOCR performs detection and recognition of text regions in the image, returning a list of recognized text strings. These strings are concatenated with spaces to produce

a single OCR-extracted document representing what a human viewer would see when examining the job posting webpage.

The OCR process captures various visual text elements including:

- Job title and company name as rendered in the webpage header
- Body text from the description, requirements, and benefits sections
- Any additional UI elements, disclaimers, or footer text
- Potential obfuscated or hidden text that might be present in fraudulent postings

For screenshots where the image file is missing or corrupted, we assign an empty OCR text string to avoid processing errors.

b) Semantic Similarity Computation: We employ Sentence-BERT (specifically the 'all-MiniLM-L6-v2' model) to encode both the database text (concatenated from the seven text fields) and the OCR-extracted text into 384-dimensional dense vector representations. Sentence-BERT is a modification of BERT that uses siamese and triplet network structures to derive semantically meaningful sentence embeddings that can be compared using cosine similarity.

For each job posting, we compute:

$$\text{similarity_score} = \frac{\text{emb}_{\text{database}} \cdot \text{emb}_{\text{OCR}}}{\|\text{emb}_{\text{database}}\| \|\text{emb}_{\text{OCR}}\|} \quad (4)$$

where $\text{emb}_{\text{database}}$ and emb_{OCR} are the Sentence-BERT embeddings of the database text and OCR text respectively. The resulting similarity score ranges from -1 (completely opposite) to 1 (perfectly aligned). For cases where OCR extraction fails or returns empty text, we assign a similarity score of 0.0.

c) Fraud Detection Hypothesis: The semantic consistency score captures several fraud indicators:

- **Content Manipulation:** Fraudulent postings may display different information on the webpage than what is stored in the database, potentially due to malicious HTML/CSS injection, client-side JavaScript manipulation, or intentional deception.
- **Professional Presentation:** Legitimate postings typically exhibit high text-image consistency, as professional companies ensure their database records match rendered content. Low consistency may indicate hastily constructed fraudulent postings with mismatched information.
- **Obfuscation Techniques:** Fraudsters might embed hidden text, use unusual formatting, or include extra persuasive/deceptive language in the visual presentation that isn't captured in structured database fields.
- **OCR Quality Proxy:** Poorly designed fraudulent webpages may have rendering issues that result in low OCR extraction quality, leading to decreased similarity scores.

2) Progressive Feature Set Construction: We systematically evaluate four feature configurations to isolate the contribution of each modality:

a) Option 1: Similarity Baseline: This configuration uses only the single-dimensional OCR similarity score as input

to classifiers. This minimal feature set tests whether text-image consistency alone provides discriminative power for fraud detection. The feature vector is simply:

$$\mathbf{x}_1 = [\text{similarity_score}] \in \mathbb{R}^1 \quad (5)$$

b) Option 2: Similarity + Text Embeddings: This configuration augments the similarity score with rich semantic text representations:

$$\mathbf{x}_2 = [\text{similarity_score}, \text{emb}_{\text{database}}] \in \mathbb{R}^{385} \quad (6)$$

where $\text{emb}_{\text{database}}$ is the 384-dimensional Sentence-BERT embedding of the concatenated database text. This combination allows classifiers to learn from both semantic content (what the posting says) and structural consistency (whether it matches its visual representation). The text embeddings capture semantic patterns common in fraudulent language, such as urgency, vague job descriptions, or unrealistic promises.

c) Option 3: Similarity + Image Embeddings: This configuration replaces text embeddings with visual features:

$$\mathbf{x}_3 = [\text{similarity_score}, \text{emb}_{\text{image}}] \in \mathbb{R}^{513} \quad (7)$$

where $\text{emb}_{\text{image}}$ is a 512-dimensional feature vector extracted from the webpage screenshot. We use a pre-trained ResNet18 model (trained on ImageNet) as a feature extractor. Specifically, we remove the final fully-connected classification layer and use the global average pooling output from the last convolutional block, which provides a rich representation of visual characteristics.

Image preprocessing consists of:

- Resize to 224×224 pixels (ResNet18's expected input size)
- Convert to tensor and normalize with ImageNet statistics
- For missing or corrupted images, use a zero vector of dimension 512

The ResNet18 features capture visual design elements that may correlate with fraud:

- Layout professionalism and consistency
- Color schemes and visual branding quality
- Presence of logos, images, or graphical elements
- Text density and spatial organization
- Overall visual complexity and polish

d) Option 4: Full Multimodal Integration: This configuration combines all three information sources:

$$\mathbf{x}_4 = [\text{similarity_score}, \text{emb}_{\text{database}}, \text{emb}_{\text{image}}] \in \mathbb{R}^{897} \quad (8)$$

This 897-dimensional feature vector (1 + 384 + 512) represents the richest feature space, allowing classifiers to learn complex interactions between semantic content, visual appearance, and text-image consistency. The hypothesis is that combining all modalities will provide the most robust fraud detection, as different types of fraud may be more evident in different modalities.

3) *Feature Caching and Computational Efficiency*: To enable rapid experimentation, we implement a caching system for expensive feature computations:

- **Text Embeddings Cache**: Sentence-BERT encodings of database text are computed once and saved to `text_embs_500.npy`. Subsequent runs load from cache rather than re-encoding.
- **Image Embeddings Cache**: ResNet18 features are similarly cached in `img_embs_500.npy` after first computation.
- **OCR Results**: While not explicitly cached in the provided code, OCR extraction could be cached to avoid redundant text recognition across experiments.

This caching strategy dramatically reduces experimental iteration time, as feature extraction (especially OCR and CNN forward passes) is the computational bottleneck. Once features are cached, training multiple classical ML models takes only seconds.

4) *Classical Machine Learning Classifiers*: For each of the four feature configurations, we train and evaluate two complementary classical ML algorithms:

a) *Logistic Regression*: We employ L2-regularized logistic regression as a linear baseline:

- **Solver**: liblinear (efficient for relatively small datasets)
- **Regularization**: L2 penalty with default strength ($C=1.0$)
- **Maximum Iterations**: 2000 (ensures convergence even for high-dimensional features)
- **Feature Scaling**: StandardScaler (zero mean, unit variance) applied to all features

Logistic regression provides interpretable linear decision boundaries and serves as a strong baseline for many classification tasks. Its coefficients can be analyzed to understand which features (or feature dimensions) are most predictive of fraud. The feature scaling is critical for logistic regression, as features span different scales (similarity score $[0,1]$, embeddings $\mathbb{R}^{384}$ or $\mathbb{R}^{512}$).

b) *Random Forest*: We employ an ensemble of decision trees to capture non-linear patterns:

- **Number of Estimators**: 200 trees (balancing performance and computational cost)
- **Split Criterion**: Gini impurity (default for classification)
- **Tree Depth**: Unlimited (trees grow until leaves are pure or contain minimum samples)
- **Feature Scaling**: Not applied (decision trees are invariant to monotonic transformations)

Random forests provide several advantages: (1) they can learn complex non-linear decision boundaries and feature interactions, (2) they are robust to outliers and don't require feature scaling, (3) they provide built-in feature importance metrics through mean decrease in impurity, and (4) they naturally handle high-dimensional feature spaces through random feature subsampling at each split.

The contrast between logistic regression (linear, parametric) and random forest (non-linear, non-parametric) allows us to

assess whether the multimodal features exhibit primarily linear or non-linear relationships with fraud labels.

5) *Training and Evaluation Protocol*: For all experiments in Approach 3, we follow a consistent protocol:

- **Dataset**: 500-row random sample from EMSCAD (`random_state=42` for reproducibility)
- **Train-Test Split**: Stratified 80-20 split maintaining class distribution (`random_state=42`)
- **Feature Scaling**: Applied only to logistic regression inputs using StandardScaler fitted on training data
- **Model Training**: Fit classifiers on training features/labels
- **Prediction**: Generate class predictions and probability estimates on test set
- **Model Persistence**: Save trained models using joblib for reproducibility and deployment

The stratified splitting is crucial given the severe class imbalance, ensuring both training and test sets contain proportional representation of fraudulent and legitimate postings. This prevents scenarios where the minority class might be entirely absent from one partition.

E. Computational Considerations and Implementation Details

All three approaches were implemented in Python using PyTorch (deep learning), scikit-learn (classical ML and preprocessing), transformers (BERT), EasyOCR (OCR), and sentence-transformers (Sentence-BERT). Experiments were conducted on [specify hardware: e.g., NVIDIA Tesla V100 GPU with 16GB memory, Intel Xeon CPU, 64GB RAM].

For Approaches 1 and 2, the 20% dataset subsampling was necessary due to memory constraints and training time considerations. The full EMSCAD dataset contains over 17,000 job postings; processing all webpage screenshots through deep CNN models and BERT would require substantial computational resources. The subsampling strategy with fixed random seed ensures reproducibility while making experiments tractable.

For Approach 3, we demonstrate the methodology's feasibility with minimal computational requirements. The feature caching system enables rapid iteration over multiple ML algorithms and hyperparameter configurations without repeatedly performing expensive OCR and CNN inference.

V. RESULTS AND ANALYSIS

We evaluate our three multimodal approaches on the EMSCAD dataset augmented with Selenium-captured webpage screenshots using stratified 80-20 train-test splits (3,576 test samples: 3,403 legitimate, 173 fraudulent).

A. Overall Performance Comparison

Table I presents performance metrics across all approaches:

B. Key Findings

Best Performance: The BERT + Multi-Kernel ResNet + CV approach achieves highest metrics (98.8% accuracy, 0.8814 F1, 82.08% recall), demonstrating that the combination

TABLE I
OVERALL PERFORMANCE COMPARISON

Approach	Acc	Prec	Recall	F1	AUC
Baseline [1]	0.9885	—	—	0.88	—
BERT + ResNet	0.9863	0.9366	0.7688	0.8444	0.9743
BERT + Multi-Kernel ResNet + CV	0.9871	0.9045	0.8208	0.8606	0.9758
OCR-Based Classical ML	0.94	0.80	0.74	0.85	0.78

of multi-kernel convolutions with traditional CV features (TF-IDF, BoW) provides superior discriminative capability.

Multimodal Advantage: All approaches significantly outperform the text-only baseline (0.88 F1), with improvements of +0.8 to +4.1 F1 points, validating the importance of visual information integration.

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